Anchor Method Revision Notes

Gated Floater-Aware Anchoring and Dense-Force Projection Risk

Revision package for *Wind-Driven Deformable 3D Gaussian Splatting: Idea Sketch*

Prepared for repo-level idea-sketch revision

May 11, 2026

Core update. This revision keeps the fixed-topology Gaussian-aware material anchoring design, but strengthens two details. First, isolation is no longer used as an independent Gaussian reliability factor; it becomes a **gated floater likelihood** that strongly penalizes a Gaussian only when it is simultaneously isolated, low-opacity, far from the proxy surface, and not protected by boundary, thin-structure, constraint, or view evidence. Second, because Blender dense-force supervision will be available, the load-sample score restores a **dense-force-supervised projection-risk field** for conservation-preserving dense-to-proxy force projection.

1 Purpose and Intended Use

This document is a repo-ready update note for revising the current idea sketch. It should be placed next to the existing sketch and given to a VSCode coding/writing agent as the technical specification for the anchor-method update.

The original sketch already frames the central loop as

$$\text{3DGS} \rightarrow \text{simulation proxy} \rightarrow \text{wind dynamics} \rightarrow \text{Gaussian attribute transport} \rightarrow \text{3DGS rendering.}$$

The current document keeps that structure, but makes the anchor construction more precise and more defensible.

1.1 Most important correction

The previous phrasing can be misread in two problematic ways:

1. **Raw Gaussian count problem.** A region with many Gaussians is not necessarily physically important. It may reflect texture complexity, specular artifacts, view coverage imbalance, floaters, or reconstruction noise.
2. **Frame-wise wind exposure problem.** Wind exposure can change every frame, especially under wind-field simulation. However, the method should not change anchors or remesh every frame.

The corrected design is:

> **Anchor placement is computed once in material space. Runtime updates only states and force values.**
> Raw Gaussian density is replaced by reliability-weighted Gaussian transport demand. Frame-wise exposure is evaluated on fixed load samples and projected to persistent simulation anchors.

2 Final Recommended Method Name

The preferred compact method name is:

Gaussian-aware material anchoring

The preferred full technical name is:

Reliability-weighted Gaussian-aware fixed-topology material anchor mesh with fixed material-space load sampling

Use the compact name in titles and section headings. Use the full name in the method definition or contribution paragraph.

3 Expression Candidates to Preserve

The following expression candidates should be preserved in the repo notes. Some are method names, some are terminology variants, and some are safe replacement phrases.

3.1 Primary method-name candidates

- Fixed-topology Gaussian-aware material anchor mesh
- Gaussian-aware fixed-topology material anchor mesh
- Persistent Gaussian-Aware Material Anchors
- Gaussian-aware persistent simulation anchors
- Wind-aware Gaussian-to-proxy material anchoring
- Mesh-derived simulation anchors
- Physics anchors / mesh-proxy anchors
- Simulation node / physics node
- Adaptive Simulation Mesh Anchoring, with explicit note: preprocessing-only
- 3DGS-derived adaptive simulation anchor mesh, with explicit note: preprocessing-only
- Reliability-weighted Gaussian-aware material anchoring
- Reliability-weighted Gaussian transport demand
- Gated floater likelihood
- Thin-structure-protected floater suppression
- Projection-risk-aware load sampling
- Dense-force-supervised load-sample placement
- Conservation-aware force projection risk
- Fixed material-space load sampling

3.2 Anchor type terms

- **Simulation anchors**: persistent vertices of the simulation proxy, $A_{\text{sim}} = V_{\text{sim}}$.
- **Constraint anchors**: pinned, support, or user-selected subsets of simulation anchors.
- **Load samples / load anchors**: fixed material-space quadrature sites used to sample wind exposure and aerodynamic loads.
- **Binding anchors**: persistent triangle-local references used for Gaussian binding.
- Material-space anchors.
- Persistent material anchors.
- Time-varying anchor states.

3.3 Pipeline/runtime terms

- one-time simulation-oriented proxy construction
- fixed-topology mesh-proxy simulation
- persistent Gaussian-to-mesh binding
- triangle-local Gaussian binding
- mesh-local frame transport
- Gaussian attribute transport
- Gaussian-opacity-based wind exposure
- canonical exposure-sensitivity prior
- fixed material-space load samples
- conservation-preserving dense-to-proxy force projection
- dense-force-supervised projection-risk field
- gated floater likelihood
- projection-risk-aware load-sample placement
- deformation-aware SH residual correction
- optional runtime LOD over fixed anchors
- active-anchor LOD
- constraint LOD

3.4 Phrases to avoid and replacements

Avoid	Use instead
adaptive remeshing during simulation	one-time simulation-oriented remeshing
dynamic anchor adjustment	time-varying anchor states
per-frame proxy reconstruction	fixed-topology runtime simulation
time-varying Gaussian binding	persistent Gaussian-to-mesh binding
direct Gaussian-patch aerodynamics	mesh-proxy wind load projection
AI SH coefficient prediction	deformation-aware SH residual correction
Gaussian-dense regions are important	reliability-weighted Gaussian transport demand
place anchors where frame-wise exposure changes	fixed canonical exposure prior plus runtime load sampling

4 Core Runtime Invariant

The method must explicitly state that remeshing and Gaussian rebinding are not runtime operations.

```
Preprocessing / setup:
  - construct proxy surface or mesh
  - compute scalar fields for resolution design
  - remesh once into a fixed-topology simulation proxy
  - choose persistent simulation anchors
  - place fixed material-space load samples
  - bind Gaussians to proxy triangles once
  - construct constraint graph once

Runtime:
  - update x_i(t), v_i(t) for simulation anchors
  - update exposure and force values at fixed load samples
  - project load-sample forces to simulation anchors
  - solve XPBD/PBD on fixed topology
  - update mesh-local frames
  - transport Gaussian mean, covariance, and appearance residual

Not performed at runtime:
  - remeshing
  - Gaussian rebinding
  - proxy reconstruction
  - topology change
```

Formally,

$$V_{\text{sim}}, E_{\text{sim}}, F_{\text{sim}}, \mathcal{C}, \mathcal{B}_{G\to M}, \mathcal{Q}_{\text{load}} \quad \text{are fixed after setup,}$$

while

$$x_i(t), v_i(t), e_l(t), f_l(t), R_T(t), \mu_j(t), \Sigma_j(t), c_j(t) \quad \text{are runtime variables.}$$

5 Separate Simulation Anchors from Load Samples

This is the central structural correction.

5.1 Simulation anchors

Simulation anchors are mesh vertices:

$$A_{\text{sim}} = V_{\text{sim}}.$$

Each anchor carries a physical state:

$$a_i(t) = (x_i(t), v_i(t), m_i, \theta_i),$$

where $x_i(t)$ is position, $v_i(t)$ is velocity, m_i is mass, and θ_i is a material or scene-calibrated parameter vector.

A typical mass assignment is

$$m_i = \rho A_i, \qquad A_i = \sum_{T \ni i} \frac{\text{area}(T)}{3}.$$

5.2 Load samples

Load samples are fixed material-space quadrature sites on faces. A load sample l is stored as

$$q_l = (T_l, b_l),$$

where T_l is a triangle id and b_l is a barycentric coordinate. Runtime position is

$$x_l(t) = \sum_{i \in T_l} b_{li} x_i(t).$$

Exposure is evaluated at load samples, not by reselection of anchors:

$$e_l(t) = \text{sample}\left(S_t, \pi_w(x_l(t))\right).$$

The effective relative wind can be written as

$$u_l^{\text{eff}}(t) = e_l(t) W(x_l(t), t) - v_l(t),$$

where

$$v_l(t) = \sum_{i \in T_l} b_{li} v_i(t).$$

The predicted or analytic load at the load sample is

$$f_l(t) = P_\theta\left(z_l(t), u_l^{\text{eff}}(t), e_l(t), \theta_l\right).$$

Finally, the load is projected to simulation anchors by barycentric accumulation:

$$f_i(t) \mathrel{+}= b_{li} f_l(t), \qquad i \in T_l.$$

5.3 Why this resolves the exposure objection

Wind exposure may change every frame:

$$e_l(t) \neq e_l(t+1).$$

This does not imply anchor changes because the material-space support is fixed:

$$q_l = (T_l, b_l) \quad \text{is fixed.}$$

Thus the method supports frame-wise wind fields while maintaining persistent simulation anchors and persistent Gaussian bindings.

6 Revised Score Fields

Use two resolution-design fields: one for simulation anchors and one for load samples.

6.1 Simulation-anchor score

$$s_{\text{sim}}(p) = w_B B(p) + w_C C_{\text{constraint}}(p) + w_K K_{\text{geom}}(p) + w_P P_{\text{phys}}(p) + w_R R_G(p).$$

This score is used to determine target simulation edge length:

$$h_{\text{sim}}(p) = \operatorname{clamp}\left(\frac{h_0}{\sqrt{1 + s_{\text{sim}}(p)}}, h_{\min}, h_{\max} \right).$$

6.2 Load-sample score

Because Blender simulation will provide dense force supervision, the full load-sample score should include a projection-risk field:

$$s_{\text{load}}(p) = w_E E_{\text{prior}}(p) + w_F F_{\text{projection\_risk}}(p) + w_R^{\text{load}} R_G(p) + w_K^{\text{load}} K_{\text{geom}}(p).$$

This score controls load-sample density, not the solver mesh topology:

$$h_{\text{load}}(p) = \operatorname{clamp}\left(\frac{h_{0,\text{load}}}{\sqrt{1 + s_{\text{load}}(p)}}, h_{\text{min,load}}, h_{\text{max,load}} \right).$$

The projection-risk term should primarily affect fixed load-sample density. If a coarse simulation proxy cannot represent the corrected load distribution even after denser load sampling, a weak version may optionally be fed into the simulation-anchor score:

$$s_{\text{sim}}(p) \leftarrow s_{\text{sim}}(p) + \lambda_{F\to S} F_{\text{projection\_risk}}(p), \qquad \lambda_{F\to S} \ll w_F.$$

6.3 Recommended priority

Priority	Term	Reason
1	$B(p)$, $C_{\text{constraint}}(p)$	Boundary, pin, support, and user controls must be preserved.
2	$K_{\text{geom}}(p)$	Thin structures, curvature, folds, tips, and silhouettes must not collapse.
3	$P_{\text{phys}}(p)$	Regions expected to strain or bend strongly need adequate simulation resolution.
4	$F_{\text{projection\_risk}}(p)$	For load samples, Blender dense-force supervision can identify where coarse projection would require large force/wrench corrections.
5	$R_G(p)$	Stable Gaussian-to-mesh binding and attribute transport require enough proxy resolution.
6	$E_{\text{prior}}(p)$	Exposure sensitivity is mainly for load-sample density, not simulation-anchor density.
7	Raw Gaussian density	Do not use directly.

7 Term-by-Term Computation

This section gives concrete formulas for each score term.

7.1 $R_G(p)$: reliability-weighted Gaussian transport demand

7.1.1 Definition and role

$R_G(p)$ is not physical importance and not raw Gaussian count. It estimates where the proxy needs enough resolution to bind and transport reliable, surface-consistent, rendering-contributing Gaussians.

Let a trained Gaussian be

$$g_j = (\mu_j, R_j, s_j, \alpha_j, c_j),$$

where μ_j is mean, R_j is local orientation, s_j is scale, α_j is opacity, and c_j is the SH appearance parameter. Because c_j is already used for SH, do not use c_j as a scalar contribution weight in this score. Use η_j instead.

Project each Gaussian to the initial proxy surface M:

$$p_j = \arg\min_{p\in M} \|\mu_j - p\|, \qquad d_j = \|\mu_j - p_j\|.$$

Define

$$R_G(p) = 1 - \exp\left(-\frac{\widetilde{R}_G(p)}{\tau_R}\right),$$

with

$$\widetilde{R}_G(p) = \sum_j q_j \eta_j K_j(p).$$

A robust automatic choice is

$$\tau_R = P_{90}(\widetilde{R}_G),$$

where P_{90} is the 90th percentile over mesh vertices or faces.

7.1.2 Surface kernel $K_j(p)$

MVP isotropic kernel:

$$K_j(p) = \exp\left(-\frac{\|p - p_j\|^2}{2r_j^2}\right),$$

with

$$r_j = \text{clamp}\left(2\max_k s_{j,k}, r_{\min}, r_{\max}\right).$$

More accurate tangent-plane version:

$$\Sigma_j = R_j \,\text{diag}(s_j^2) R_j^\top,$$

$$\Sigma_j^{\text{tan}} = T_j^\top \Sigma_j T_j,$$

where $T_j = [t_{j,1}, t_{j,2}]$ is a local surface tangent basis at p_j. Then

$$K_j(p) = \exp\left(-\frac{1}{2}(u - u_j)^\top (\Sigma_j^{\text{tan}} + \epsilon I)^{-1}(u - u_j)\right).$$

For the first implementation, the isotropic version is sufficient.

7.1.3 Gaussian reliability q_j: gated floater likelihood

Important revision. Do not use isolation as an independent reliability term. Legitimate sparse structures such as leaf tips, antenna-like details, thin stems, or sparse but valid cloth edges may be isolated in the Gaussian set. Isolation should only contribute to a gated floater likelihood.

Base reliability without isolation. Use a geometric mean of positive reliability factors, excluding isolation:

$$q_j^{\text{base}} = \exp\left(\frac{\sum_m \lambda_m \log(q_j^m + \epsilon)}{\sum_m \lambda_m}\right),$$

where the active factors can include

$$q_j^m \in \{q_j^\alpha, q_j^{\text{surf}}, q_j^{\text{normal}}, q_j^{\text{view}}\}.$$

If normal or view terms are not implemented yet, set them to 1 rather than replacing them with an isolation term.

Opacity reliability.

$$q_j^\alpha = \text{smoothstep}(P_{20}(\alpha), P_{80}(\alpha), \alpha_j).$$

Percentile thresholds are preferred over fixed thresholds because Gaussian opacity scales vary across assets.

Surface consistency.

$$q_j^{\text{surf}} = \exp\left(-\frac{d_j^2}{2\sigma_d^2}\right).$$

A practical default is

$$\sigma_d = h_0 \quad \text{or} \quad \sigma_d = 2\,\text{median}_j(\max_k s_{j,k}).$$

Normal consistency. Let n_j^G be the Gaussian normal candidate, e.g., the eigenvector of the smallest Gaussian scale, and let $n_j^M = n_M(p_j)$ be the mesh normal. Then

$$q_j^{\text{normal}} = |\langle n_j^G, n_j^M \rangle|^\gamma, \qquad \gamma \approx 2.$$

If the Gaussian normal estimate is unstable, set $q_j^{\text{normal}} = 1$ for the MVP.

View contribution reliability. If renderer instrumentation is available, accumulate alpha-compositing contribution over training views:

$$A_{j,v}(u) = T_{j,v}(u)\alpha_j K_{j,v}(u),$$

$$C_j = \sum_v \sum_u A_{j,v}(u),$$

$$q_j^{\text{view}} = \text{normalize}\left(\log(1 + C_j)\right).$$

If renderer instrumentation is not available, approximate by visible camera count:

$$q_j^{\text{view}} = \min\left(1, \frac{N_j^{\text{visible}}}{N_0}\right), \qquad N_0 \in [3, 5].$$

For the MVP, this term can be omitted by setting $q_j^{\text{view}} = 1$.

Isolation as a floater-likelihood gate, not as reliability. Let

$$N_j^{\text{nbr}} = \#\{k \mid \|\mu_k - \mu_j\| < r_{\text{nbr}}\}.$$

Define isolatedness as

$$I_j = \exp\left(-\frac{N_j^{\text{nbr}}}{N_{\text{ref}}}\right), \qquad N_{\text{ref}} \approx 6.$$

Thus $I_j \approx 1$ means the Gaussian is isolated, while $I_j \approx 0$ means it has enough neighbors.

Define low-opacity and surface-far scores:

$$L_j = 1 - q_j^{\alpha}, \qquad D_j = 1 - q_j^{\text{surf}}.$$

Then define a protection score for valid sparse detail:

$$S_j = \max\left(B(p_j), K_{\text{geom}}(p_j), C_{\text{constraint}}(p_j), T_{\text{thin}}(p_j), q_j^{\text{view}}\right),$$

where $T_{\text{thin}}(p_j)$ is an optional thin-structure prior. If it is not implemented, use

$$S_j = \max\left(B(p_j), K_{\text{geom}}(p_j), C_{\text{constraint}}(p_j), q_j^{\text{view}}\right).$$

The gated floater likelihood is

$$p_j^{\text{float}} = I_j L_j D_j (1 - S_j).$$

A Gaussian is therefore strongly penalized only when it is simultaneously isolated, low-opacity, far from the proxy surface, and not protected by boundary, curvature, thin-structure, constraint, or view-contribution evidence.

The final reliability becomes

$$q_j = q_j^{\text{base}} \exp\left(-\lambda_{\text{float}} p_j^{\text{float}}\right), \qquad \lambda_{\text{float}} \in [1, 2].$$

A clipped linear alternative is also acceptable:

$$q_j = q_j^{\text{base}} \operatorname{clamp}(1 - \lambda_{\text{float}} p_j^{\text{float}}, 0, 1).$$

Optional Gaussian dropping rule. Do not drop a Gaussian simply because it is isolated. A safer dropping condition is

$$\text{drop}(g_j) = [p_j^{\text{float}} > \tau_{\text{float}}] \wedge [q_j^{\alpha} < \tau_{\alpha}] \wedge [q_j^{\text{surf}} < \tau_{\text{surf}}] \wedge [S_j < \tau_S].$$

This protects legitimate sparse thin structures while still removing obvious low-opacity, far-from-surface floaters.

7.1.4 Rendering contribution η_j

If C_j is available,

$$\eta_j = \text{normalize}(\log(1 + C_j)).$$

If not, use the simplest defensible prototype setting:

$$\eta_j = 1,$$

and put all reliability filtering into q_j.

7.1.5 MVP version of R_G

For initial development, still avoid direct q_j^{iso} multiplication. Use the gated simplification:

$$q_j^{\text{base}} = \left(q_j^{\alpha} q_j^{\text{surf}}\right)^{1/2},$$

$$I_j = \exp\left(-\frac{N_j^{\text{nbr}}}{6}\right), \qquad S_j = \max(B(p_j), K_{\text{geom}}(p_j), C_{\text{constraint}}(p_j)),$$

$$p_j^{\text{float}} = I_j(1 - q_j^{\alpha})(1 - q_j^{\text{surf}})(1 - S_j),$$

$$q_j = q_j^{\text{base}} \exp(-\lambda_{\text{float}} p_j^{\text{float}}).$$

Then compute

$$R_G(p) = 1 - \exp\left(-\frac{\sum_j q_j K_j(p)}{P_{90}(\sum_j q_j K_j(p))}\right).$$

This avoids raw Gaussian count and avoids killing legitimate sparse thin structures merely because they are isolated.

7.2 $B(p)$: boundary / pinned / free edge prior

$B(p)$ preserves free boundaries, pinned boundaries, and support edges. Detect boundary edges by incident face count:

$$e \in \partial M \quad \Longleftrightarrow \quad |\mathcal{F}(e)| = 1.$$

Let $\mathcal{L}_{\text{bdry}}$ be the set of boundary loops. Then

$$B(p) = \max_{\ell \in \mathcal{L}_{\text{bdry}}} w_\ell \exp\left(-\frac{d_{\text{geo}}(p,\ell)^2}{2r_B^2}\right).$$

Recommended default:

$$r_B = 2h_0.$$

Example boundary weights:

$$w_{\text{pinned}} = 1.0, \qquad w_{\text{free}} = 0.7, \qquad w_{\text{ordinary}} = 0.5.$$

Pinned/support curves should be hard-preserved by the remesher, not merely upweighted:

$$A_{\text{pin}} \subset V_{\text{sim}}.$$

7.3 $K_{\text{geom}}(p)$: curvature / thin-structure prior

This term preserves folds, leaf tips, thin edges, sharp silhouettes, and narrow strips.

Cotangent mean-curvature version at vertex i:

$$H_i n_i = \frac{1}{2A_i} \sum_{j \in \mathcal{N}(i)} (\cot \alpha_{ij} + \cot \beta_{ij})(x_i - x_j),$$

$$\kappa_i = \|H_i n_i\|.$$

Simpler face-normal variation version:

$$K_{\text{geom}}(f) = \frac{1}{|\mathcal{N}(f)|} \sum_{f' \in \mathcal{N}(f)} \frac{1 - \langle n_f, n_{f'} \rangle}{\|c_f - c_{f'}\| + \epsilon}.$$

Use robust percentile normalization:

$$\widehat{K}_{\text{geom}}(p) = \text{clamp}\left(\frac{\kappa(p) - P_{50}(\kappa)}{P_{95}(\kappa) - P_{50}(\kappa) + \epsilon}, 0, 1\right).$$

For the prototype, the face-normal variation version is recommended because it is easier to implement and less sensitive to mesh operators.

7.4 $P_{\text{phys}}(p)$: expected physical deformation prior

This term estimates where strain or bending is expected to be high. It should not be invented as an arbitrary weight. Use either of the following.

7.4.1 MVP choice

Set

$$P_{\text{phys}}(p) = 0.$$

This is acceptable for a first implementation.

7.4.2 Full paper choice: one-time pilot simulation

Run a cheap preprocessing-only pilot simulation over sampled canonical wind directions $\mathcal{W}$. For edge $e = (i, k)$, define stretch response:

$$s_e(t) = \frac{\big|\|x_i(t) - x_k(t)\| - \|x_i^0 - x_k^0\|\big|}{\|x_i^0 - x_k^0\| + \epsilon}.$$

For a bending edge, define dihedral response:

$$b_e(t) = |\theta_e(t) - \theta_e^0|.$$

Accumulate a surface prior:

$$P_{\text{phys}}(p) = \text{normalize}\left(\max_{w\in\mathcal{W}} \max_t \left[s(p, t, w) + \lambda_b b(p, t, w)\right]\right).$$

This is still preprocessing only and does not violate fixed-topology runtime.

7.5 $E_{\text{prior}}(p)$: canonical wind-exposure sensitivity prior

E_{prior} is not a frame-wise anchor update signal. It is a rest-pose prior used mainly to place fixed load samples.

Choose a family of possible wind directions:

$$\mathcal{W} = \{w_1, \dots, w_N\}.$$

For each wind direction, compute rest-pose Gaussian-opacity transmittance:

$$S_w^0(u) \approx \prod_j \left(1 - \alpha_j K_j^w(u)\right).$$

Then sample exposure on surface point p:

$$e_w^0(p) = \text{sample}(S_w^0, \pi_w(p)).$$

Two useful sensitivity measures are exposure variance and exposure gradient:

$$E_{\text{var}}(p) = \text{Var}_{w\in\mathcal{W}}[e^0_w(p)],$$

$$E_{\text{grad}}(i) = \max_{w\in\mathcal{W}} \max_{k\in\mathcal{N}(i)} \frac{|e^0_w(i) - e^0_w(k)|}{\|x_i - x_k\| + \epsilon}.$$

Combine and normalize:

$$E_{\text{prior}}(p) = \text{normalize}\left(E_{\text{var}}(p) + \lambda_g E_{\text{grad}}(p)\right), \qquad \lambda_g \in [0.5, 1.0].$$

Recommended use:

$$E_{\text{prior}}(p) \quad \text{strongly affects } s_{\text{load}}(p), \quad \text{weakly or not at all affects } s_{\text{sim}}(p).$$

7.6 $F_{\text{projection\_risk}}(p)$: dense-force-supervised projection risk

Because the data pipeline will obtain simulation supervision from Blender, the paper can use the stronger dense-force-supervised version of projection risk. This term connects the anchor/load sampling design to the existing contribution candidate: conservation-preserving dense-to-proxy force projection.

7.6.1 Dense force input from Blender

Assume Blender or another simulator provides high-resolution surface samples r with dense forces f_r^w under canonical wind direction or condition $w \in \mathcal{W}$. Partition the surface into local patches Ω_m, such as proxy faces, geodesic clusters, or load-sampling cells. For a patch Ω, define the dense net force and dense wrench:

$$F^{\text{dense}}_{\Omega,w} = \sum_{r\in\Omega} f_r^w,$$

$$T^{\text{dense}}_{\Omega,w} = \sum_{r\in\Omega} (x_r - c_\Omega) \times f_r^w,$$

where c_Ω is the patch centroid or the center of mass of dense samples in the patch.

7.6.2 Raw dense-to-proxy projection

Let $i \in \mathcal{I}(\Omega)$ be proxy anchors or load samples associated with patch Ω. A raw projected force can be obtained by nearest projection, barycentric projection, or area-weighted accumulation:

$$F^{\text{raw}}_{i,w} = \sum_{r\in\Omega} a_{ri} f_r^w, \qquad \sum_{i\in\mathcal{I}(\Omega)} a_{ri} = 1.$$

If torque samples are used, define $\tau^{\text{raw}}_{i,w}$ analogously. For a first implementation, local torques can be omitted and all wrench conservation can be handled by corrected forces; the full paper version can include torque variables.

7.6.3 Conservation-preserving correction

Compute corrected proxy forces by solving

$$\min_{\{F^*_{i,w}\}} \sum_{i\in\mathcal{I}(\Omega)} \|F^*_{i,w} - F^{\text{raw}}_{i,w}\|^2$$

subject to

$$\sum_{i\in\mathcal{I}(\Omega)} F^*_{i,w} = F^{\text{dense}}_{\Omega,w},$$

$$\sum_{i\in\mathcal{I}(\Omega)} (x_i - c_\Omega) \times F^*_{i,w} = T^{\text{dense}}_{\Omega,w}.$$

The torque-augmented version is

$$\min_{\{F^*_{i,w},\tau^*_{i,w}\}} \sum_i \|F^*_{i,w} - F^{\text{raw}}_{i,w}\|^2 + \gamma \sum_i \|\tau^*_{i,w} - \tau^{\text{raw}}_{i,w}\|^2,$$

subject to

$$\sum_i F^*_{i,w} = F^{\text{dense}}_{\Omega,w},$$

$$\sum_i (x_i - c_\Omega) \times F^*_{i,w} + \sum_i \tau^*_{i,w} = T^{\text{dense}}_{\Omega,w}.$$

This gives both a corrected training target and a diagnostic signal for whether the load sampling is too coarse.

7.6.4 Projection-risk field

Define the patch-level correction ratio:

$$\epsilon^{\text{proj}}_{\Omega,w} = \frac{\sum_i \|F^*_{i,w} - F^{\text{raw}}_{i,w}\|^2 + \gamma \sum_i \|\tau^*_{i,w} - \tau^{\text{raw}}_{i,w}\|^2}{\sum_i \|F^{\text{raw}}_{i,w}\|^2 + \gamma \sum_i \|\tau^{\text{raw}}_{i,w}\|^2 + \epsilon}.$$

If the no-torque version is used, drop the torque terms. The surface field is

$$F_{\text{projection\_risk}}(p) = \text{normalize}\left(\max_{w\in\mathcal{W}} \epsilon^{\text{proj}}_{\Omega(p),w}\right).$$

A high value means that coarse load sampling would require a large correction to preserve net force and wrench; therefore, the region should receive denser fixed material-space load samples.

7.6.5 Use in scores

The recommended full load-score is

$$s_{\text{load}}(p) = 1.5E_{\text{prior}}(p) + 1.2F_{\text{projection\_risk}}(p) + 0.7R_G(p) + 0.5K_{\text{geom}}(p).$$

For a Blender-supervised MVP where E_{prior} is not yet implemented, use

$$s^{\text{Blender-MVP}}_{\text{load}}(p) = 1.2F_{\text{projection\_risk}}(p) + 0.5K_{\text{geom}}(p).$$

Only feed this field weakly into s_{sim} if the corrected load distribution cannot be represented by the current simulation proxy:

$$s_{\text{sim}}(p) \leftarrow s_{\text{sim}}(p) + \lambda_{F\to S} F_{\text{projection\_risk}}(p), \qquad \lambda_{F\to S} \in [0.2, 0.4].$$

7.7 $C_{\text{constraint}}(p)$: user / artist controllability prior

Let $\mathcal{H} = \{h_m\}$ be user handle curves, support regions, pin points, flag-pole edges, curtain rails, leaf stems, or other controllability regions. Then

$$C_{\text{constraint}}(p) = \max_m \exp\left(-\frac{d_{\text{geo}}(p,h_m)^2}{2r_m^2}\right).$$

The exact handle or pin positions should be hard-preserved:

$$h_m \subset V_{\text{sim}} \quad \text{or} \quad d(h_m, V_{\text{sim}}) < \epsilon_h.$$

8 Practical Weights

All fields should be normalized to $[0, 1]$ before combination:

$$\widehat{F}(p) = \text{clamp}\left(\frac{F(p) - P_5(F)}{P_{95}(F) - P_5(F) + \epsilon}, 0, 1\right).$$

8.1 Recommended full score

$s_{\text{sim}}(p) = 2.0B(p) + 2.0C_{\text{constraint}}(p) + 1.5K_{\text{geom}}(p) + 1.0P_{\text{phys}}(p) + 0.7R_G(p) + \lambda_{F\to S}F_{\text{projection\_risk}}(p)$, with $\lambda_{F\to S} = 0$ by default and $\lambda_{F\to S} \in [0.2, 0.4]$ only when the proxy cannot represent corrected loads.

$$s_{\text{load}}(p) = 1.5E_{\text{prior}}(p) + 1.2F_{\text{projection\_risk}}(p) + 0.7R_G(p) + 0.5K_{\text{geom}}(p).$$

8.2 Recommended MVP score

For first implementation:

$$s_{\text{sim}}(p) = 2.0B(p) + 1.5K_{\text{geom}}(p) + 0.7R_G(p).$$

If Blender dense forces are available before E_{prior} is implemented, use

$$s_{\text{load}}^{\text{Blender-MVP}}(p) = 1.2F_{\text{projection\_risk}}(p) + 0.5K_{\text{geom}}(p).$$

Otherwise use

$$s_{\text{load}}^{\text{MVP}}(p) = 1.5E_{\text{prior}}(p) + 0.5K_{\text{geom}}(p),$$

or uniform face quadrature until exposure-sensitive and projection-risk-aware load sampling are implemented.

9 Gaussian-to-Mesh Binding and Transport

For Gaussian g_j bound to triangle $T_j = (a, b, c)$, store

$$\mathcal{B}_j = (T_j, b_j, R_{T_j}^0, \delta_j),$$

where $b_j = (b_{ja}, b_{jb}, b_{jc})$ is barycentric coordinate, $R_{T_j}^0$ is the rest local frame, and δ_j is an optional local offset.

Runtime mean transport:

$$\mu_j(t) = b_{ja}x_a(t) + b_{jb}x_b(t) + b_{jc}x_c(t) + R_{T_j}(t)\delta_j.$$

Covariance transport:

$$\Sigma_j(t) = R_{T_j}(t)\Sigma_j^0 R_{T_j}(t)^\top.$$

Appearance correction:

$$c_j(t) = R_{\text{SH}}(\Delta T_j(t))c_j^0 + \Delta c_j(t).$$

The key requirement is persistence:

$$T_j, b_j, \delta_j \quad \text{are not recomputed every frame.}$$

10 Implementation Pseudo-Pipeline

```
Input:
  trained 3DGS G
  initial proxy mesh M_raw
  optional user pin / handle regions
  expected wind direction set W

Step 1. Gaussian reliability
  for each Gaussian g_j:
    p_j = nearest point on M_raw to mu_j
    d_j = ||mu_j - p_j||
    q_alpha = smoothstep(P20(alpha), P80(alpha), alpha_j)
    q_surf = exp(-d_j^2 / (2 * sigma_d^2))
    q_normal = abs(dot(n_G_j, n_M(p_j)))^gamma # optional
    q_view = visible-view or renderer contribution score # optional
    q_base = geometric_mean(q_alpha, q_surf, q_normal, q_view)
    I = exp(-N_nbr_j / 6) # isolatedness, not reliability
    S = max(B(p_j), K_geom(p_j), C_constraint(p_j), q_view)
    p_float = I * (1 - q_alpha) * (1 - q_surf) * (1 - S)
    q_j = q_base * exp(-lambda_float * p_float)

Step 2. Gaussian transport demand
  initialize R_raw on mesh vertices or faces
  for each Gaussian g_j:
    splat q_j * eta_j to nearby surface points using K_j(p)
  R_G = 1 - exp(-R_raw / percentile90(R_raw))

Step 3. Geometry and boundary fields
  K_geom = robust-normalized curvature or face-normal variation
  B = geodesic distance field to boundary / pinned / free edges
  C_constraint = distance field to handles and support regions

Step 4. Optional fields
  P_phys = one-time pilot simulation strain/bending response
  E_prior = rest-pose exposure variance/gradient over sampled wind
      directions
  F_projection_risk = Blender dense-force projection correction ratio

Step 5. Resolution fields
  s_sim = 2.0B + 2.0C + 1.5K_geom + 1.0P_phys + 0.7R_G
  s_load = 1.5E_prior + 1.2F_projection_risk + 0.7R_G + 0.5K_geom
  h_sim(p) = clamp(h0 / sqrt(1 + s_sim(p)), h_min, h_max)
  h_load(p) = clamp(h0_load / sqrt(1 + s_load(p)), h_load_min, h_load_max)

Step 6. One-time construction
  remesh once with h_sim while preserving pins and boundaries
  set simulation anchors A_sim = V_sim
  place fixed load samples using h_load
  bind each Gaussian to a triangle id + barycentric coordinate + local
      frame

Runtime:
  no remeshing
  no Gaussian rebinding
  evaluate exposure and loads at fixed load samples
  use Blender-corrected dense-to-proxy force targets when training
```

```
project load forces to fixed simulation anchors while preserving net
    force and wrench
solve XPBD/PBD on fixed topology
transport Gaussian mean, covariance, and appearance residual
```

11 Paste-Ready Paper Text

11.1 Contribution statement

We introduce a reliability-weighted Gaussian-aware fixed-topology material anchor mesh that serves as a persistent simulation scaffold for 3D Gaussian assets. The proxy resolution is not determined by raw Gaussian density. Instead, it is guided by boundary preservation, user constraints, thin-structure geometry, optional pilot-simulation response, and reliability-weighted Gaussian transport demand with a gated floater likelihood. Since Blender provides dense force supervision, we further place fixed material-space load samples using a dense-force-supervised projection-risk field that estimates where coarse projection would require large correction to preserve net force and wrench. Runtime wind exposure is evaluated on fixed load samples and projected to persistent simulation anchors, enabling force-space wind dynamics without frame-wise remeshing or Gaussian rebinding.

11.2 Method paragraph

Our method constructs the simulation proxy once in the canonical configuration. During runtime, the proxy topology, simulation anchors, constraint graph, load-sample material coordinates, and Gaussian-to-mesh bindings remain fixed. Only anchor states, wind exposure values, external loads, local frames, and Gaussian attributes are updated. This design avoids frame-wise remeshing and rebinding, which are expensive and can introduce temporal popping.

11.3 Clarification paragraph for Gaussian density

We do not treat raw Gaussian density as physical importance. A dense Gaussian region may result from texture complexity, specular effects, view-coverage bias, or reconstruction noise. Instead, we compute a reliability-weighted Gaussian transport demand by splatting surface-consistent and optionally rendering-contributing Gaussians onto the proxy surface. Isolation is not used as an independent rejection factor; it contributes only to a gated floater likelihood that activates when a Gaussian is isolated, low-opacity, far from the proxy surface, and not protected by boundary, curvature, thin-structure, constraint, or view evidence. This field indicates where the proxy requires sufficient resolution for stable Gaussian-to-mesh binding and attribute transport.

11.4 Clarification paragraph for wind exposure

We do not change anchors according to frame-wise exposure. A canonical exposure-sensitivity prior can be precomputed from a sampled family of wind directions and used only for fixed load-sample placement. During runtime, exposure values are evaluated on these fixed material-space load samples and projected as forces to persistent simulation anchors.

11.5 Clarification paragraph for gated floater likelihood

We do not use isolation as an independent Gaussian reliability factor because legitimate thin structures may be sparsely represented. Instead, isolation participates only in a gated floater likelihood. A Gaussian is strongly penalized only when it is simultaneously isolated, low-opacity, far from the proxy surface, and not protected by boundary, curvature, thin-structure, constraint, or view-contribution evidence.

11.6 Clarification paragraph for Blender dense-force projection risk

Since Blender provides dense force supervision, we estimate a projection-risk field by measuring how much a raw dense-to-proxy force projection must be corrected to preserve patch-level net force and wrench. High-risk regions receive denser fixed material-space load samples. This keeps the conservation-preserving dense-to-proxy force projection contribution connected to the anchor and load-sampling design.

11.7 Korean summary paragraph

본방법은 raw Gaussian 개수를물리적중요도로보지않는다. 대신 opacity, surface consistency, optional normal/view contribution, 그리고 gated floater likelihood 를반영한 reliability-weighted Gaussian transport demand 를계산하여 Gaussian attribute transport 에 필요한 proxy 해상도를정한다. Isolation 은단독제거조건으로쓰지않고, low opacity, far from surface, isolated, not protected 조건이함께나타날때만강한 penalty 로사용한다. 또한 wind exposure 가매프레임바뀌더라도 anchor 를재배치하지않는다. Simulation anchor 는 fixed-topology mesh vertex 로유지하고, wind exposure 와 load 는 fixed material-space load sample 에서매프레임평가한뒤 simulation anchor 로 projection 한다. Blender dense force supervision 을사용할경우, net force 와 wrench 보존을위해필요한 correction 크기를 projection risk 로계산하고, 이값을 load sample density 에반영한다.

12 How to Update the Existing Idea Sketch

12.1 Section 5: Method Sketch

Add after the proxy-remeshing step:

> The remeshing is performed only once during preprocessing. The resulting simulation proxy has fixed topology during animation. Its vertices become persistent simulation anchors. We separately place fixed material-space load samples for wind exposure and force evaluation.

Replace any statement that says “Gaussian density” with:

> reliability-weighted Gaussian transport demand.

12.2 Section 8: Adaptation Novelty

Add a new novelty item:

> **Reliability-weighted Gaussian-aware material anchoring.** The proxy resolution is not based on raw Gaussian count. Instead, the method splats reliability-weighted Gaussian transport demand, boundary and constraint priors, thin-structure geometry, and optional pilot physical response onto the proxy surface. Simulation anchors and load samples are separated: anchors define fixed-topology physical states, while fixed load samples evaluate frame-varying wind exposure.

12.3 Section 9: Mathematical Formulation

Insert the following formulas:

$$s_{\text{sim}}(p) = w_B B(p) + w_C C_{\text{constraint}}(p) + w_K K_{\text{geom}}(p) + w_P P_{\text{phys}}(p) + w_R R_G(p),$$
$$s_{\text{load}}(p) = w_E E_{\text{prior}}(p) + w_F F_{\text{projection\_risk}}(p) + w_R^{\text{load}} R_G(p) + w_K^{\text{load}} K_{\text{geom}}(p).$$

Then define R_G, gated floater likelihood, B, K_{geom}, P_{phys}, E_{prior}, $F_{\text{projection\_risk}}$, and $C_{\text{constraint}}$ using the term-by-term equations in this document.

12.4 Section 10: Formula Transfer Notes

Add:

> Adapted: raw Gaussian coverage is replaced by reliability-weighted Gaussian transport demand with gated floater likelihood. Runtime wind exposure is not used to remesh or reselect anchors; it is evaluated at fixed material-space load samples and projected to persistent simulation anchors. Blender dense force supervision is used to compute a projection-risk field for load-sample placement and conservation-preserving dense-to-proxy force targets.

12.5 Section 15: Ablation Ideas

Add the following ablations:

- raw Gaussian density vs. reliability-weighted Gaussian transport demand
- uniform simulation anchors vs. Gaussian-aware material anchors
- joint anchor/load sampling vs. separated simulation anchors and load samples
- fixed load samples vs. per-frame exposure-driven resampling
- persistent Gaussian binding vs. per-frame rebinding
- direct isolation reliability vs. gated floater likelihood
- no floater gate vs. gated floater likelihood with thin-structure protection
- no canonical exposure-sensitivity prior vs. exposure-aware load sampling
- no projection-risk term vs. dense-force-supervised projection-risk load sampling
- raw dense-to-proxy force projection vs. conservation-preserving force/wrench projection
- MVP score vs. full score with pilot physical prior

12.6 Section 16: Risks and Failure Cases

Add:

- Raw Gaussian density may reflect scanning artifacts, texture, view imbalance, or floaters rather than physical importance. Mitigation: use reliability-weighted transport demand with saturation.
- Frame-varying exposure may be misread as requiring frame-varying anchors. Mitigation: keep simulation anchors fixed and evaluate exposure at fixed material-space load samples.

- Isolation-based reliability may suppress valid sparse thin structures. Mitigation: use isolation only inside a gated floater likelihood with boundary, curvature, thin-structure, constraint, and view-contribution protection.
- If conservation-preserving force projection is not connected to sampling design, the contribution may look disconnected. Mitigation: add $F_{\text{projection\_risk}}$ to the fixed load-sample score and show an ablation.
- Reliability terms may become over-engineered. Mitigation: provide an MVP version using opacity, surface distance, and gated floater likelihood.

13 Decision Log Entry to Add

2026-05-11 Anchor Construction Update
Decision. Replace raw Gaussian-density-based anchor placement with reliability-weighted Gaussian transport demand, and separate persistent simulation anchors from fixed material-space load samples. Simulation anchors remain fixed-topology mesh vertices. Load samples evaluate frame-varying wind exposure and project forces to simulation anchors.
Reason. Raw Gaussian count can reflect capture artifacts, texture, specular effects, or floaters rather than physical importance. Wind exposure changes over time under wind simulation, but changing anchors or remeshing every frame would be too expensive and would break persistent Gaussian binding.
Next. Implement the MVP score $s_{\text{sim}} = 2.0B + 1.5K_{\text{geom}} + 0.7R_G$, with R_G computed from opacity, surface-distance, and isolation reliability. Add fixed load samples and later extend them with canonical exposure-sensitivity prior.

14 Additional Decision Log Entry

2026-05-11 Gated Floater and Projection-Risk Update
Decision. Replace direct isolation reliability with a gated floater likelihood. Restore a dense-force-supervised projection-risk field in the fixed load-sample score because Blender simulation data will provide dense force supervision.
Reason. Isolated Gaussians are not always floaters; legitimate sparse thin structures such as leaf tips, stems, cloth edges, or antenna-like details may be isolated but valid. A Gaussian should be strongly penalized only when isolation co-occurs with low opacity, large surface distance, and no boundary/thin-structure/view protection. Also, conservation-preserving dense-to-proxy force projection is a stated contribution; using Blender dense forces to compute projection risk connects that contribution to load-sample design.
Next. Implement gated floater likelihood in $R_G(p)$, compute $F_{\text{projection\_risk}}(p)$ from Blender dense force correction ratios, and update s_{load} to include $w_F F_{\text{projection\_risk}}(p)$.

15 VSCode / Coding-Agent Prompt

```
Read docs/anchor_method_revision_notes_2026-05-11.pdf and update the
    current Wind-Driven Deformable 3D Gaussian Splatting idea sketch
    accordingly.

Revision goals:
1. Keep the original project framing:
   3DGS input -> simulation proxy -> wind dynamics -> Gaussian attribute
      transport -> 3DGS rendering.
```

```
2. Use the recommended method name:
   Reliability-weighted Gaussian-aware fixed-topology material anchor mesh
   or the shorter term Gaussian-aware material anchoring.
3. Clarify that the method performs no per-frame remeshing, no per-frame
    proxy reconstruction, and no per-frame Gaussian rebinding.
4. Define simulation anchors as persistent mesh vertices: A_sim = V_sim.
5. Separate simulation anchors from fixed material-space load samples.
6. Replace raw Gaussian density with reliability-weighted Gaussian
    transport demand R_G(p).
7. Define R_G(p) explicitly from Gaussian projection, surface kernel K_j(
    p), reliability q_j, optional rendering contribution eta_j, and
    saturation.
8. Define q_j from opacity reliability, surface consistency, optional
    normal consistency, optional view contribution, and gated floater
    likelihood. Do not multiply by q_iso as an independent reliability
    factor.
9. Define gated floater likelihood p_float = I * L * D * (1 - S), where
    isolation is penalized only when combined with low opacity, far-from-
    surface evidence, and no thin/boundary/view protection.
10. Define two score fields:
   s_sim(p) for simulation proxy resolution,
   s_load(p) for fixed load-sample density.
11. Because Blender dense forces are available, add F_projection_risk(p)
    to s_load(p) and define it from the correction ratio required for
    conservation-preserving dense-to-proxy force/wrench projection.
12. Explain that canonical exposure-sensitivity prior E_prior(p) is
    preprocessing-only and mainly controls load-sample placement, not
    runtime anchor changes.
13. Insert the paste-ready contribution, method, Gaussian-density
    clarification, wind-exposure clarification, gated-floater
    clarification, and projection-risk clarification paragraphs.
14. Add the 2026-05-11 decision log entries.
15. Add ablations for raw Gaussian density vs. reliability-weighted
    demand, direct isolation reliability vs. gated floater likelihood,
    uniform anchors vs. Gaussian-aware anchors, joint anchor/load sampling
     vs. separated sampling, no projection-risk vs. projection-risk-aware
    load sampling, and persistent binding vs. per-frame rebinding.

Do not reframe the paper as an AI SH coefficient prediction paper. SH
    residual correction remains a supporting component.
```

16 Minimal Development Checklist

1. Project each Gaussian center to the initial proxy surface.
2. Compute q_j^{α}, q_j^{surf}, optional q_j^{view}, isolatedness I_j, protection score S_j, and gated floater likelihood p_j^{float}.
3. Compute $q_j = q_j^{\text{base}} \exp(-\lambda_{\text{float}} p_j^{\text{float}})$ and splat $q_j K_j(p)$ to mesh vertices or faces.
4. Saturate to get $R_G(p)$.
5. Compute boundary field $B(p)$.
6. Compute face-normal variation $K_{\text{geom}}(p)$.
7. Build MVP score $s_{\text{sim}}(p) = 2.0B(p) + 1.5K_{\text{geom}}(p) + 0.7R_G(p)$.

8. Remesh once using $h_{\text{sim}}(p)$ and preserve pinned/support boundaries.
9. From Blender dense forces, compute $F_{\text{projection\_risk}}(p)$ by measuring conservation correction ratios.
10. Place fixed load samples as (T_l, b_l) using s_{load} with the projection-risk term.
11. Bind Gaussians once as $(T_j, b_j, R^0_{T_j}, \delta_j)$.
12. Runtime: evaluate exposure and loads at load samples, project to anchors, solve XPBD/PBD, transport Gaussian attributes.

17 Final One-Line Summary

Use a fixed-topology Gaussian-aware material anchor mesh, but do not base it on raw Gaussian density or direct isolation penalties. Compute reliability-weighted Gaussian transport demand with a gated floater likelihood, keep simulation anchors fixed, and use Blender dense-force-supervised projection risk to place fixed material-space load samples for conservation-preserving wind-force projection.